

Error Codes:-

(i) Parity Method for Error Detection:-

(i) In this method a parity bit is used to detect an error.

(ii) An error is detected by using a parity bit to a group of bits which may contain an even or odd number of 1's.

(iii) A given system can only operate with even or odd parity but not both.

for instance:

a check is made to the group of bits given to a system containing even number of 1's if system operates with even parity.

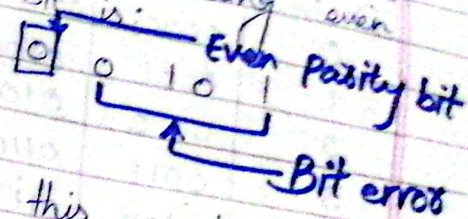
Following table illustrate the parity bits for each BCD number including both even and odd parities.

Even parity			Odd parity		
Decimal	P	BCD	P	BCD	
0	0	0000	1	0000	
1	1	0001	0	0001	
2	1	0010	0	0010	
3	0	0011	1	0011	
4	1	0100	0	0100	
5	0	0101	1	0101	
6	0	0110	1	0110	
7	1	0111	0	0111	
8	1	1000	0	1000	
9	0	1001	1	1001	

Detecting an Error:-

⇒ A parity bit provides for the detection of a single bit error but can't check for two errors in one group.

⇒ Suppose we want to transmit the BCD code 0101. The total code transmitted including even parity bit is:



When this code is received, the parity check circuitry determines that there is only a single 1 (odd number). BCD (an even number of 1s does not appear in the code when it is received, so an error is detected. Similarly, happens for odd parity.

Exers-3 code :-

Decimal	BCD	EXC-3C
0	0000	0011
1	0001	0100
2	0010	0101
3	0011	0110
4	0100	0111
5	0101	1000
6	0110	1001
7	0111	1010
8	1000	1011
9	1001	1100

Q#	Difference b/w BCD & binary code	Binary code
i)	BCD is a coding system.	- BCD is a number system.
ii)	Its base is 4	- Its base is 2.
iii)	It needs more bit to represent a number.	- It needs fewer bits to represent number.
iv)	It is easy to construct	- It is difficult to construct
e.g.	- BCD of 15 is 00010101	- Binary of 15 is 1111

Cyclic Redundancy Code

=> The cyclic redundancy is widely used code for detecting one and two bit transmission errors when digital data are transferred on a communication link.

=> CRC is often described mathematically as the division of two polynomials to generate remainder.

=> It is correct if remainder = 0.

=> It is incorrect if remainder $\neq 0$.

=> It is based on polynomials. | Modulo-2 Operations

Univariate polynomial:

When the coefficients are limited to 1s and 0s

it is called univariate polynomials.

e.g. $1x^3 + 0x^2 + 1x^1 + 1x^0$
or simply $x^3 + x^1 + x^0$

Polynomial:-

A polynomial is a mathematical expression that is a sum of terms with positive exponents.

Modulo-2 Operations:-

CRC is based on division of two binary numbers:
Division is just a series of subtraction and shifts.

Modulo-2 (Exclusive OR)

input bits		o.b
1	1	0
1	0	1
0	1	1
0	0	0

CRC Process:-

The process is as follows:
=> Select a generator code: it can have fewer bits than the data bits to be checked.
=> Append a number of 0s equal to the number of bits in the generator code to the data bits.
=> Divide the data bits including the appended bits by the generator code bits using modulo-2.
=> if the remainder is 0, the data and appended bits are sent as is.
=> if the remainder is not 0, the appended bits are made equal to the remainder bits in order to get a 0 remainder before data are sent.

$\Rightarrow$ At the receiving end, the received divided the incoming appended data bit code by the same generator code as used by the sender.
 $\Rightarrow$ if the remainder is 0, there is no error detected.

Example:- Determine the transmitted CRC for the following byte of data and generator code. Verify that the remainder is 0.

D: 11010011

G: 1010

Since the generator code has four bits, add four 0s to the data. The appended data is D'.

D' = 110100110000

Divide the appended data by using modulo-2 operation until all bits have been used.

$$\frac{D'}{G} = \frac{110100110000}{1010}$$

$$\begin{array}{r}
 110100110000 \\
 1010 \downarrow \\
 \hline
 1110 \\
 1010 \downarrow \\
 \hline
 1000 \\
 1010 \downarrow \\
 \hline
 1011 \\
 1010 \downarrow \\
 \hline
 0000 \\
 1010 \downarrow \\
 \hline
 0100
 \end{array}$$

Remainder $\neq 0$

Again append the data using remainder.

D' = 110100110100

Again Divide using Modulo-2 operation.

$$\begin{array}{r}
 110100110100 \\
 1010 \downarrow \\
 \hline
 1110 \\
 1010 \downarrow \\
 \hline
 1000
 \end{array}$$

$$\begin{array}{r}
 1000 \\
 1010 \\
 \hline
 1011 \\
 1010 \\
 \hline
 1010 \\
 1010 \\
 \hline
 0
 \end{array}$$

Remainder = 0

Hamming Code

- Hamming code was invented by R.W. Hamming code in 1950. represented as (n, k) code
- Hamming code detect 1-bit and 2-bit errors.
- it used to correct and detect errors in the transmitted data.

Encoding data message such as:

- Selecting the number of redundant bits
 - Selecting after choosing the location of redundant bits
 - Assigning the value to redundant bit
- $\Rightarrow n$ is total bit and k is message bit

Use parity bit = 2^n $\{n=0, 1, 2, \dots\}$

Example:

Encode a binary number into even parity HC.
Message: 1001

	P_4	P_2	P_1	
0	0	0	0	1
1	0	0	1	0
2	0	1	1	0
3	0	1	0	0
4	1	0	1	1
5	1	0	0	0
6	1	1	1	0
7	1	1	0	0

7-bit hamming code

$P_1 = D_3, D_5, D_7$

$P_2 = D_3, D_6, D_7$

$P_4 = D_5, D_6, D_7$

$P_1 = D_3 \oplus D_5 \oplus D_7$

$0 = 1 \oplus 0 \oplus 1$

$P_2 = D_3 \oplus D_6 \oplus D_7$

$0 = 1 \oplus 0 \oplus 1$

$P_4 = D_5 \oplus D_6 \oplus D_7$

$1 = 0 \oplus 0 \oplus 1$

generated = 1001100

After change bit let's

1101100

Now, there is error.

Let's check

1	1	1	1	1	0	0
---	---	---	---	---	---	---

D₇ D₆ D₅ P₄ D₃ P₂ P₁

$$P_1 = D_3, D_5, D_7$$

$$0 = 1 \oplus 0 \oplus 1 \quad P_1 = 0 \text{ (correct)}$$

$$P_2 = D_3, D_6, D_7$$

$$1 = 1 \oplus 1 \oplus 1 \quad P_2 = 1 \text{ (error)}$$

$$P_3 = D_5 \oplus D_6 \oplus D_7$$

$$1 = 1 \oplus 1 \oplus 1 \quad P_4 = 1 \text{ (error)}$$

Error Correction

$$P_4 P_2 P_1 = 110 = 6$$

in 6-bit there is error if
0 then 1 or if 1 then zero

1	1	1	1	1	0	0
---	---	---	---	---	---	---

1	0	1	1	1	0	0
---	---	---	---	---	---	---

Application of HC

- Computers memory
- Satellites
- Modems
- Plasma CAM
- Shielding wire
- Open connectors

Q#:- if received hamming code is 1110101 with even parity then detect ^{and} correct _{error}

7	6	5	4	3	2	1
D ₇	D ₆	D ₅	P ₄	D ₃	P ₂	P ₁

Parity bit = $2^n \{0, 1, 2, \dots\}$

D ₇	D ₆	D ₅	P ₄	D ₃	P ₂	P ₁
----------------	----------------	----------------	----------------	----------------	----------------	----------------

$$P_1 = D_7 \oplus D_5 \oplus D_3$$

D ₇	D ₆	D ₅	P ₄	D ₃	P ₂	P ₁
----------------	----------------	----------------	----------------	----------------	----------------	----------------

$$P_2 = D_3 \oplus D_6 \oplus D_7$$

D ₇	D ₆	D ₅	P ₄	D ₃	P ₂	P ₁
----------------	----------------	----------------	----------------	----------------	----------------	----------------

$$P_4 = D_5 \oplus D_6 \oplus D_7$$

D ₇	D ₆	D ₅	P ₄	D ₃	P ₂	P ₁
1	1	1	0	1	0	1

$$P_1 = D_3 \oplus D_5 \oplus D_7$$

$$= 1 \oplus 1 \oplus 1$$

$$1 = 0 \oplus 1$$

$$1 = 1 \} - P_1 = 0 \quad \checkmark$$

$$P_2 = D_3 \oplus D_6 \oplus D_7$$

$$0 = 1 \oplus 1 \oplus 1$$

$$= 0 \oplus 1$$

$$0 = 1 \} \rightarrow P_2 = 1 \quad \times$$

$$P_4 = D_1 \oplus D_6 \oplus D_5$$

$$= 1 \oplus 1 \oplus 1$$

$$0 = 0 \oplus 1$$

$$0 = 1 \} - P_4 = 1 \quad \times$$

Location = $P_4 P_2 P_1 = 110 = 6$ bit
on 6-bit error.

$$\text{Error syndrome} = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\text{Received data} = \begin{bmatrix} 1 & 1 & 1 & 0 & 1 & 0 & 1 \end{bmatrix}$$

$$\text{Corrected data} = R \oplus E$$

$$= \begin{array}{ccccccc} \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \\ \oplus \\ \begin{bmatrix} 1 & 1 & 1 & 0 & 1 & 0 & 1 \end{bmatrix} \end{array}$$

$$C.D = \begin{bmatrix} 1 & 0 & 1 & 0 & 1 & 0 & 1 \end{bmatrix}$$

$$0 = 1 \oplus 0 \oplus 1$$

$$P_1 = 0 \text{ (Correct)}$$

Generation of Hamming code

Q# 5-bit data 01101 is given.
Present given data in HC.

→ $K = 5$ bits

→ identify P.B = $2^P \geq P + K + 1$

$$\Rightarrow 2^P \geq P + 5 + 1$$

$$\Rightarrow 2^P \geq P + 6$$

for $P = 1$

$$2^1 \geq 1 + 6 \Rightarrow 2 \geq 7 \text{ (false)}$$

for $P = 2$

$$2^2 \geq 2 + 6 \Rightarrow 4 \geq 8 \text{ (false)}$$

for $P = 3$

$$2^3 \geq 3 + 6 \Rightarrow 8 \geq 9 \text{ (false)}$$

for $P = 4$

$$2^4 \geq 4 + 6 \Rightarrow 16 \geq 10 \text{ (True)}$$

So, we have 4 parity bits.

⇒ So for $P=4, K=5, n=9$
⇒ This is (9, 5) hamming code

9	8	7	6	5	4	3	2	1
D ₄	P ₈	D ₇	D ₆	D ₅	P ₄	D ₃	P ₂	P ₁

→ Position of P.B | value of P.B

$$P_1 = 2^0 = 1$$

$$P_2 = 2^1 = 2$$

$$P_3 = 2^2 = 4$$

$$P_8 = 2^3 = 8$$

$$P_1 \rightarrow D_3 D_5 D_7 D_9$$

$$P_2 \rightarrow D_3 D_6 D_7$$

$$P_3 \rightarrow D_5 D_6 D_7$$

$$P_8 \rightarrow D_7$$

0	0	1	1	0	0	1	1	0
---	---	---	---	---	---	---	---	---

$$\rightarrow P_1 = D_3 \oplus D_5 \oplus D_7 \oplus D_9$$

$$P_1 = 1 \oplus 0 \oplus 1 \oplus 0 = 0$$

$$\rightarrow P_2 = D_3 \oplus D_6 \oplus D_7$$

$$P_2 = 1 \oplus 1 \oplus 1 = 1$$

$$\rightarrow P_4 = D_5 \oplus D_6 \oplus D_7$$

$$P_4 = 0 \oplus 1 \oplus 1 = 0$$

$$\rightarrow P_8 = 0$$

Addition Using

Exc-3

if carry = 0 then - 0011
 if carry = 1 then + 0011

Example:-

(1)₁₀ $\xrightarrow{\text{BCD}}$ 0001 $\xrightarrow{\text{Exc-3}}$ 0100
 (2)₁₀ $\xrightarrow{\text{BCD}}$ 0010 $\xrightarrow{\text{Exc-3}}$ 0101

Adding :-

0	1	0	0	
+	0	1	0	1
carry = 0	1	0	0	1
Subtract - 0011	0	0	1	1
	0	1	1	0 (correct)

1 + 3 = 4
 2 + 3 = 5
 3 + 6 = 9
 = 6

Example #2:-

(9)₁₀ $\xrightarrow{\text{BCD}}$ 1001 $\xrightarrow{\text{Exc-3}}$ 1100
 (3)₁₀ $\xrightarrow{\text{BCD}}$ 0011 $\xrightarrow{\text{Exc-3}}$ 0110
 Adding

$$\begin{array}{r}
 1100 \\
 + 0110 \\
 \hline
 000\boxed{1}0010
 \end{array}$$

Add 0011

$$\begin{array}{r}
 00010010 \\
 + 00110011 \\
 \hline
 (01000101) \text{ Ans}
 \end{array}$$

Example :-

$$\begin{array}{lcl}
 (12)_{10} & \xrightarrow{\text{BCD}} & 00010010 \xrightarrow{\text{Exc-3}} 01000101 \\
 (13)_{10} & \xrightarrow{\text{BCD}} & 00010011 \xrightarrow{\text{Exc-3}} 01000110
 \end{array}$$

Adding :-

$$\begin{array}{r}
 01000101 \\
 + 01000110 \\
 \hline
 1.0001011 \\
 00110011 \\
 \hline
 01011000
 \end{array}$$

subtract-0011

Verification :-

$$\begin{array}{r}
 12 \\
 + 13 \\
 \hline
 25 \\
 + 33 \\
 \hline
 58
 \end{array}$$

2421 Code

Decimal	2421 code
0	0000
1	0001
2	0010
3	0011
4	0100
5	1001
6	1100
7	1101
8	1110
9	1111

After 5 we obtain Taking 2's complement of 1 to 5 digit.

=> +ively weighted code.

=> Unnatural code. Sum of weights of unnatural Bcd code is equal to 9.

=> Also called reflective code

84-2-1 Code

Decimal 84-2-1

0	0000	9	1111
1	0111		
2	0110		
3	0101		
4	0100		
5	1011		
6	1010		
7	1001		
8	1000		

=> Negatively weighted code.

=> Unnatural Bcd code.

=> it is self-complementing code.

Parity Generators/Checkers

⇒ A parity bit indicates if the number of 1's in a code is even or odd for the purpose of error detection.

⇒ In order to check for or to generate the proper parity in a given code, a basic principle can be used.

The sum of even number of 1s is always 0 and the sum of odd number of 1s is always 1.

What is parity generator?

Parity generator:

⇒ A parity generator is a combinational logic circuit that generates parity bit in the transmitter.

⇒ A parity bit is used for the purpose of detecting errors during transmission of binary information.

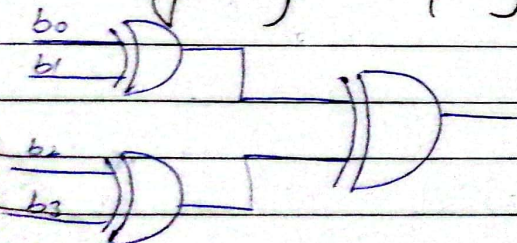
=> It is an extra bit included with a binary message to make the number of 1's either odd or even.

4-bit even parity generator:

The circuit for generating parity bits and checking the parity bits of a given word can be designed using gates.

=> EX-OR gate are ideal for checking the parity of a binary number because they produce an output when the input has odd number of ones.

Circuit diagram of even parity generator:-



	b_3	b_2	b_1	b_0	E.P	O.P
0	0	0	0	0	0	1
1	0	0	0	1	1	0
2	0	0	1	0	1	0
3	0	0	1	1	0	1
4	0	1	0	0	1	0
5	0	1	0	1	0	1
6	0	1	1	0	0	1
7	0	1	1	1	1	0
8	1	0	0	0	1	0
9	1	0	0	1	0	1
10	1	0	1	0	0	1
11	1	0	1	1	1	0
12	1	1	0	0	0	1
13	1	1	0	1	1	0
14	1	1	1	0	1	0
15	1	1	1	1	0	1

$b_3 b_2 b_1 b_0$	00	01	11	10	$b_3 b_2 b_1 b_0$	00	01	11	10
00	0 ₀	1 ₁	0 ₃	1 ₂	00	1 ₀	0 ₁	1 ₁	0 ₂
01	1 ₁	0 ₅	1 ₇	0 ₆	01	0 ₁	1 ₀	0 ₁	1 ₀
11	0 ₂	1 ₃	0 ₅	1 ₄	11	1 ₁	0 ₁	1 ₁	0 ₁
10	1 ₂	0 ₉	1 ₁	0 ₁₀	10	0 ₁	1 ₁	0 ₁	1 ₁

E.P Expression

$$P = b_0 \oplus b_1 \oplus b_2 \oplus b_3$$

O.D Parity expression

Even parity expression:-

$$\begin{aligned}
 P_e &= \bar{b}_0 \bar{b}_1 \bar{b}_2 \bar{b}_3 + \bar{b}_0 \bar{b}_1 b_2 \bar{b}_3 + \bar{b}_0 b_1 \bar{b}_2 \bar{b}_3 \\
 &+ \bar{b}_0 b_1 b_2 \bar{b}_3 + b_0 \bar{b}_1 \bar{b}_2 \bar{b}_3 + b_0 \bar{b}_1 b_2 \bar{b}_3 + b_0 b_1 \bar{b}_2 \bar{b}_3 \\
 &+ b_0 b_1 b_2 \bar{b}_3 + b_0 \bar{b}_1 b_2 b_3 + b_0 b_1 \bar{b}_2 b_3 + b_0 b_1 b_2 b_3 \\
 &= \bar{b}_3 \bar{b}_2 (\bar{b}_1 b_0 + b_1 \bar{b}_0) + \bar{b}_3 b_2 (\bar{b}_1 b_0 + b_1 \bar{b}_0) \\
 &+ b_3 \bar{b}_2 (b_0 \bar{b}_1 + b_1 \bar{b}_0) + b_3 b_2 (\bar{b}_1 \bar{b}_0 + b_1 b_0) \\
 &= \bar{b}_3 \bar{b}_2 (b_0 \oplus b_1) + \bar{b}_3 b_2 (\bar{b}_0 \oplus \bar{b}_1) \\
 &+ b_3 \bar{b}_2 (b_0 \oplus b_1) + b_3 b_2 (\bar{b}_0 \oplus \bar{b}_1) \\
 &= (b_0 \oplus b_1) (\bar{b}_3 \bar{b}_2 \oplus b_3 \bar{b}_2) + (\bar{b}_0 \oplus \bar{b}_1) (\bar{b}_3 b_2 \oplus b_3 b_2) \\
 &= (b_0 \oplus b_1) (b_2 \oplus b_2) + (\bar{b}_0 \oplus \bar{b}_1) (b_2 \oplus b_2)
 \end{aligned}$$

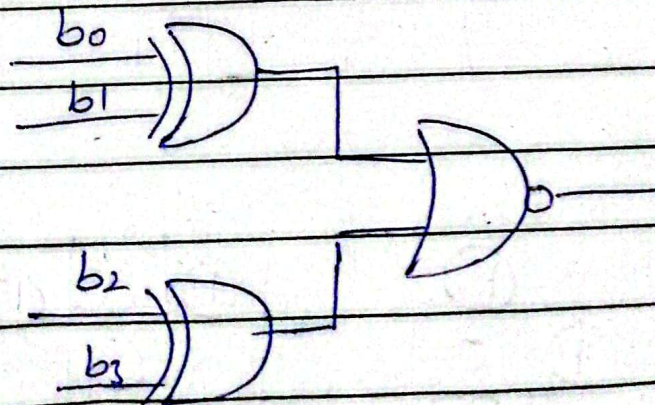
$$P_e = (b_0 \oplus b_1) \oplus (b_2 \oplus b_3)$$

Odd parity expression

$$P_o = (b_0 \oplus b_1) \oplus (b_2 \oplus b_3)$$

$$P_o = (b_0 \oplus b_1) \oplus b_2 \oplus b_3$$

Odd parity logical diagram



Note: 1) No error is represented by 0

2) Error is represented by 1

Redundant bit:

- ! Redundant bits are extra bits that are generated and added to be information-carrying bits of data transfer to ensure that no bits were lost during the data transfer.

The number of redundant bits can be calculated using the following formula.

$$2^r \geq m + r + 1$$

m = data bit

r = redundant bit

Define parity bit.

- A parity bit is a bit that is appended to a data of a binary bits to ensure that the total number of 1's in the data is even or odd.

Parity bits are used for error detection.

There are two types of parity:
even parity odd parity

Define Terms.

Single bit errors:

The change made in one bit in the entire data sequence is known as single bit error.

- This error only occurs in a parallel communication system because data is transferred bitwise in a single line.

Multiple bit errors:

- If there is a change in two or more bits of data sequence of a transmitter to receiver, it is known as multiple bit errors.

- This type of error occurs in both serial and parallel type data communication.

Burst errors:

- The change of the set of bits in data sequence is known as burst error.

- This type of error is calculated in from the first-bit change to last bit change.

What are error detection methods?

It is called 3 error detection technique.

- Simple parity check
- Two dimensional parity check
- Check sum
- Cyclic redundancy check

Which is the most efficient error correction method?

Parity check is efficient method

What is CRC Method?

- Cyclic redundancy check is an error detection method.
- It is based on binary division.

CRC generator:

- CRC generator is an algebraic polynomial represented as a bit pattern.

Types of error detection code:

- Longitudinal redundancy check
- CRC
- Parity checking